

III. Vitamins and Nutrition

Notes

Food :

- Food is any substance consumed to provide nutritional support for an organism.
- It is usually of plant or animal origin and contains essential nutrients such as carbohydrates, fats, proteins, vitamins and minerals.
- The substance is ingested by an organism and assimilated by the organism's cell to provide energy, maintain life or stimulate growth.

Components of Food :

- The major components of our food are-
 - (i) Carbohydrates
 - (ii) Fats
 - (iii) Proteins
 - (iv) Minerals (salts)
 - (v) Nucleic Acids
 - (vi) Vitamins
 - (vii) Water

Carbohydrates :

- A carbohydrate is a biomolecule consisting of carbon (C), hydrogen (H) and oxygen (O) atoms, usually with a hydrogen - oxygen atom ratio of 2 : 1 and thus with the empirical formula $C_x(H_2O)_y$ (Where x may or may not be different from y).
- Carbohydrates are the main component of food and source of immediate energy to the body.
- Approximately 55-75% energy is provided by carbohydrates to the total energy requirement of the body.
- It is stored in the liver and skeletal muscles in the form of glycogen which gives energy on requirement.
- One gram of glucose gives 4.5K calories energy on oxidation.
- **Lactose** is a natural sugar (a carbohydrate). It is milk sugar. It is responsible for the mildly sweet taste of milk.
- Lactobacillus bacteria converts lactose into lactic acid. These bacteria are employed in the manufacturing of curd and cheese.
- The **starch** and **cellulose** both are of plant origin. Starch is composed of glucose molecules linked in long chains. Cellulose is also a long chain of glucose molecules.
- The iodine test is used to test for the presence of starch. Starch turns into an intense 'blue-black' colour upon addition of aqueous solutions of the triiodide ion, due to the formation of an intermolecular charge - transfer complex.

- Cellulose does not turn into blue-black colour in the presence of iodine.

Fats :

- Fats are the compounds of Carbon, Hydrogen and Oxygen.
- In fats, the ratio of oxygen to hydrogen atom is less.
- The fats are formed with the chemical reaction of one molecule of glycerol and three molecules of fatty acids.
- Fats are the source of stored energy. Fats are broken down in the healthy body to release their constituents, glycerol and fatty acids. Glycerol itself can be converted to glucose by the liver and so become a source of energy.
- 9.0-9.3K calories energy is produced by total combustion of one gram of fat.
- It provides almost 15-20% energy to the total energy requirement of the body.
- The percentage of fat in a healthy man is about 7% to its total body weight.
- The fats have been classified into two groups on the basis of carbon atom bonds in their fatty acids–
 - (i) Saturated fats.
 - (ii) Unsaturated fats.

(i) Saturated Fats :

- A saturated fat is a type of fat in which the fatty acid chains have all or predominantly single bonds.
- Saturated fatty acids are important to nutrition because of their ability to elevate blood lipid levels in humans.
- Saturated fatty acids are not essential because the body can synthesize these fatty acids, hence nutritional recommendation around the world suggest that saturated fatty acid intake be kept low to promote health and reduce the risk from cardiovascular diseases.
- Saturated fatty acids in the diet raise plasma cholesterol and PUFA (polyunsaturated fatty acid) lowers it, decreasing the risk of heart disease.
- Following are some saturated fatty acids–
 - Butyric acid (contained in butter)
 - Lauric acid (contained in coconut oil, palm kernel oil and breast milk)
 - Myristic acid (contained in cow's milk and dairy products)
 - Palmitic acid (contained in butter, palm oil and meat)
 - Stearic acid (contained in meat and cocoa butter)

(ii) Unsaturated Fats :

- An unsaturated fat is a fat or fatty acid in which there is at least one double bond or triple bond within the fatty acid chain. A fatty acid chain is **monounsaturated** if it contains one double bond, and **polyunsaturated** if it contains more than one double bond.

- Such fatty acids are found in fish, corn, walnuts, sunflower seeds, soyabeans, cotton seeds, olive oil and sunflower oil.
- Diet high in unsaturated fatty acids and low in saturated fatty acids have been correlated with low serum cholesterol levels.
- The unsaturated fats have lower melting point and most are liquid at room temperature.
- Following are the examples of unsaturated fatty acids—
 - Palmitoleic acid (found in macadamia nuts)
 - Oleic acid (found in olive oil and canola oil)
 - Myristoleic acid (found in whale blubber)
 - Linoleic acid (found in peanut oil, chicken fat and olive oil)
 - Linolenic acid (found in linseed oil)
- Unsaturated fatty acid can not be synthesized by the body and is therefore essential to the diet. In humans, linoleic acid and α -linolenic acid are the primary essential fatty acids.
- Among oilseeds, linseed oil is with maximum quantity of α -linolenic fatty acid.
- α -Linolenic acid is an omega-3 fatty acid which prevents the accumulation of fat in blood vessels.
- Arachidonic fatty acid is a polyunsaturated omega-6 essential fatty acid. It is found in the liver, brain and glandular organs and is a constituent of animal's phosphatides. It is formed by the synthesis from dietary linoleic acid and is a precursor in the biosynthesis of prostaglandins, thromboxanes and leukotrienes while other fatty acids (non-essential) are synthesized in mammals by acetyl coenzyme A, which is formed during oxidation of glucose.
- Hydrogenation is a process in which unsaturated fats such as vegetable oils are hydrogenated to produce saturated fats, which have more desirable physical properties e.g. they melt at a desirable temperature (about 30°C to 40°C).

Trans fats :

- The majority of trans fats are artificial fats created by adding hydrogen to liquid vegetable oil to make it a solid.
- Trans fats raise bad LDL cholesterol levels and lower good HDL cholesterol levels. Eating trans fats increases the risk of developing heart disease and stroke. It is also associated with a higher risk of developing type 2 diabetes.

Proteins :

- Proteins are the essential nutrients for the human body.
- They are one of the building blocks of body tissue and can also serve as a fuel source.
- As a fuel, proteins provide 4.0K cal energy per gram.
- Amino acids are the structural units of the protein.

- There are 10 essential amino acids and 10 non-essential amino acids which are required to synthesize protein in human body.

Essential Amino Acids :

- Essential amino acids are not synthesized in human body. Human must obtain them from their diet in order to prevent protein energy malnutrition and resulting death.
- Essential amino acids include : phenylalanine, valine, threonine, tryptophan, methionine, leucine, isoleucine, lysine, histidine and arginine.
- Arginine and histidine are essential only for juveniles but not for adults because these are synthesized in adults.

(ii) Non-essential Amino Acids :

- Non-essential amino acids are such amino acids which are synthesized within the human body.
- These are alanine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, and tyrosine.
- α -Keratin is a protein, which is the main constituent of horn, nail, hoof, hair, skin etc.
- Tyrosine helps in synthesis of neurotransmitters such as adrenaline, noradrenaline and dopamine in the brain.
- In milk, water percentage is approximately 87%. Besides it contains fats, proteins, carbohydrates, calcium, potassium, so it is considered as **Complete Diet**, which is globally accepted food for all ages.
- Nursing mothers need at least 65 grams of protein per day, while in other females its requirement per day is 45-46 grams.
- Due to protein deficiency, marasmus and kwashiorkor diseases are developed in humans.

Vitamins :

- It has been observed that certain organic compounds are required in small amounts in our diet but their deficiency causes specific diseases. These compounds are called vitamins.
- Most of the vitamins cannot be synthesised in our body but plants can synthesise almost all of them, so they are considered as essential food factors.
- Vitamins are classified into two groups depending upon their solubility in water or fat.
- (i) Fat soluble vitamins :** These are vitamins A, D, E and K. They are stored in liver and adipose (fat storing) tissues.
- (ii) Water soluble vitamins :** B group vitamins and vitamin C are soluble in water. Water soluble vitamins must be supplied regularly in diet because they are readily excreted in urine and cannot be stored (except vitamin B₁₂) in our body.

Main Vitamins, Deficiency Diseases and Sources

Vitamins	Deficiency Diseases	Sources
Vitamin A (Retinol)	Night blindness, xerophthalmia, keratinization of skin infection	Carrot, milk, cheese, butter, egg, Cod liver oil
Vitamin B ₁ (Thiamin)	Beri-beri, growth retardation	Cereals, legumes, soyabeans, milk, yeast, green vegetables
Vitamin B ₂ (Riboflavin)	Cheilosis, glossitis, dermatosis, digestive disorders	Milk, egg white, liver, kidney, yeast, meat, green leafy vegetables
Vitamin B ₃ (Niacin)	Pellagra	Egg, meat, fish, milk, yeast, vegetables, mushroom
Vitamin B ₅ (Pantothenic Acid)	Paraesthesia, burning sensation	Yeast, meat, fish, egg milk, legumes
Vitamin B ₆ (Pyridoxine)	Anaemia, skin problem and muscular twisting	milk, yeast, cereals, egg, grams, yolk
Vitamin B ₇ (Vitamin-H; Biotin)	Skin disease, fall of hair	Meat, wheat, egg, vegetables, fruits
Vitamin B ₉ (Folic Acid)	Anaemia, growth retardation	Green vegetables, legumes, yeast, liver
Vitamin B ₁₂ (Cobalamin; a vitamin with cobalt)	Pernicious anaemia, abnormality in nervous system	meat, fish, egg, curd, bacteria of intestine
Vitamin C (Ascorbic Acid)	Scurvy (bleeding gums)	Amla, fruits of citrus family, tomato and green leafy vegetables
Vitamin D (Calciferol)	Rickets, osteomalacia	Cheese, fish, egg, yolk, sunlight
Vitamin E (Tocopherol)	Sterility and weaken- ing of reproductive organ and muscles	Vegetable oils, wheat, soyabean
Vitamin K (Naphtho- quinone)	Increased blood clotting time	Green leafy vegetables, bacteria of intestine

Question Bank

1. Which of the following foods will provide the nutrient for the growth of new tissues in the human body?

(a) Fruit (b) Vegetables
(c) Cheese (d) Sweets

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

Proteins are major components of human body and are more important for growth and repair of tissues. Its main sources are pulses, soyabeans, groundnut, milk, cheese, eggs, food-grains etc.

2. Consider the following statements about probiotic food:

1. The probiotic food contains live bacteria which are considered beneficial to humans.

2. Probiotic food helps in maintaining gut flora.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither nor 2

I.A.S. (Pre) 2008

Ans. (c)

Our intestines are home to about 500 million bacteria, some good and some bad. Researches suggest that the more good bacteria in your gut, the stronger your immune system will be. Probiotics are often referred to as 'good bacteria'. They help the body to absorb important vitamins and minerals including calcium, iron, chromium and vitamins A, D, E and K, just to name a few. These probiotic foods into our eating routine can enhance and improve gut flora to boost our intestinal and overall health.

3. Consider the following statements in respect of probiotics :

1. Probiotics are made of both bacteria and yeast.

2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.

3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) 1 and 3 (d) 2 and 3

I.A.S. (Pre) 2022

Ans. (c)

Probiotics are live microorganisms (bacteria and yeasts) that are intended to have health benefits when consumed or applied to the body. Probiotics may contain a variety of microorganisms. The most common are bacteria that belong to groups called Lactobacillus and Bifidobacterium. Other bacteria may also be used as probiotics, and so may yeasts such as Saccharomyces boulardii. Probiotics naturally occur in our gut. Trillions of bacteria and other microorganisms live in every person's gut. With the action of probiotics, milk and other dairy products are easier to digest. Fermentation with live bacteria results in a unique form of lactose which is easier to digest, and many probiotic bacterial strains also produce β -galactosidase, an enzyme that acts like lactase and can help to break down lactose in the gut. L.acidophilus is a probiotic bacteria that naturally occurs in the human gut and other parts of the body. This bacteria helps the digestive system break down sugars, such as lactose, into lactic acid. Hence, statement 1 and 3 are correct, while statement 2 is incorrect.

4. Nutraceuticals are products which have-

- (a) Nutrients vitamin and minerals
- (b) Nutrients protein and fatty acids
- (c) Nutrient and toxic effect
- (d) Nutrient and medicinal effect

U.P.P.C.S. (Pre) 2009

Ans. (d)

The term 'nutraceutical' is used to describe any food or part of food supplements that offers a medical or health benefit beyond simple nutrition. Such benefits many include the prevention or recurrence of the diseases.

5. Funk invented –

- (a) Vitamins
- (b) Hormones
- (c) Proteins
- (d) Enzymes

U.P.P.C.S.(Pre) 2012

Ans. (a)

Kazimierz (Casimir) Funk is generally credited with being among the first to formulate the concept of vitamins, which he called 'vital amines' or 'vitamins'. Umetaro Suzuki had in 1910 succeeded in extracting a water-soluble complex of micronutrients from rice bran and had named it 'aberic acid'.

6. Which of the following pairs is not correctly matched?

- (a) Vitamin A – Maculan
- (b) Vitamin B – Maculan
- (c) Vitamin C – James Lind
- (d) Vitamin D – Paul Muller

M.P.P.C.S. (Pre) 2017

Ans. (d)

American biochemist Elmer Verner McCollum (The name given in the question is Maculan, which is wrongly spelt) and Marguerite Davis discovered the vitamin A. McCollum also helped to discover vitamin B and vitamin D and worked out the effect of trace elements in the diet. Scottish doctor James Lind is associated with vitamin C as he found that citrus fruits cured scurvy. Swiss chemist Paul Muller is not associated with vitamin D. He discovered the insecticidal qualities and use of DDT in the control of vector diseases such as malaria and yellow fever.

7. What are vitamins?

- (a) Organic Compound
- (b) Inorganic compound
- (c) Living organism
- (d) None of these

M.P.P.C.S. (Pre) 1998

Ans. (a)

Vitamins are organic compounds and vital nutrients that an organism requires in limited amounts. An organic chemical compound (or related set of compounds) is called a vitamin when the organism cannot synthesize the compound in sufficient quantities and must be obtained through the diet.

8. Which one of the following groups of compounds is called 'accessory dietary factor'?

- (a) Fats
- (b) Hormones
- (c) Proteins
- (d) Vitamins

U.P.R.O./A.R.O. (Pre) 2014

Ans. (d)

Frederick Gowland Hopkins was the first scientist to elucidate the 'accessory food factor', the idea that food contains trace amounts of substances essential for nutrition. Accessory food factors later came to be called vitamins.

9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Most of the vitamins are not synthesized in human body.

Reason (R) : Human organs synthesize only essential micronutrients.

Select the correct answer from the codes given below:

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false
- (d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2019

Ans. (c)

Most of the vitamins cannot be synthesized in human body, so they are considered as essential food factors. For the most part, vitamins are obtained from the diet, but some are acquired by other means: for example, microorganisms in the gut flora produce vitamin K and B₇ (Biotin); and one form of vitamin D is synthesized in skin cells when they are exposed to a certain wavelength of ultraviolet light present in sunlight. Human can produce some vitamins from precursors they consume: for example, vitamin A is synthesized from beta carotene; and vitamin B₃ (Niacin) is synthesized from the amino acid tryptophan. An essential micronutrient is a nutrient required for normal physiological function that cannot be synthesized in the human body-either at all or in sufficient quantities- and thus must be obtained from a dietary source. About 30 vitamins, minerals, fatty acids and amino acids are essential micronutrients for proper functioning of human metabolism. Thus reason (R) is false.

10. Given below are two statements, one is labelled as Assertion (A) and other as Reason (R).

Assertion (A) : Human body synthesizes all the Vitamins required by it.

Reason (R) : Vitamins are essential for proper growth of the body.

Choose the correct answer from the code given below.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

U.P. R.O./A.R.O. (Mains) 2016

U.P.P.C.S. (Pre) 2021

Ans. (d)

Most of the vitamins cannot be synthesized in the human body, so they are considered as essential food factors. Vitamins are certain organic compounds required in small amounts in our diet as they are essential for proper growth of the body and their deficiency causes specific diseases. Hence, Assertion (A) is false, but Reason (R) is true.

11. Liver stores :

- (a) A, B, C and D vitamins
- (b) Only A and B vitamins
- (c) A, B and C vitamins
- (d) None of these

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

The liver stores all fat-soluble vitamins (A, D, E, and K) as well as vitamin B₁₂. The liver's storage capacity for these vitamins ensures that the body can have a sufficient supply even when dietary intake is limited. Hence, option (d) is the correct answer.

12. Which of the following facts is NOT correct about vitamins?

- (a) Some vitamins can be synthesized by intestinal microorganisms.
- (b) Some vitamins act as hormones.
- (c) Some vitamins are stored in body.
- (d) Vitamin 'K' is a water-soluble vitamin.

U.P. B.E.O. (Pre) 2019

Ans. (d)

Vitamin A, D, E and K are fat-soluble vitamins while B group vitamins and vitamin C are soluble in water. Thus, option (d) is not correct. Facts given in other options are correct about vitamins.

13. Which of the following vitamins are fat soluble?

- (a) C and E
- (b) A and C
- (c) A and D
- (d) B₁₂ and D

R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

14. Which of the following vitamins are water soluble ?

- (a) Vitamin A
- (b) Vitamin D
- (c) Vitamin K
- (d) None of these

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

See the explanation of above question.

15. Which vitamin is soluble in water?

- (a) Vitamin 'A'
- (b) Vitamin 'B'
- (c) Vitamin 'C'
- (d) Vitamin 'D'

U.P. R.O./A.R.O. (Mains) 2021

Ans. (b) & (c)

See the explanation of above question.

16. Water soluble vitamin is –

- (a) Vitamin A
- (b) Vitamin C
- (c) Vitamin D
- (d) Vitamin E

Uttarakhand P.C.S. (Pre) 2010

Ans. (b)

See the explanation of above question.

17. Which vitamins is water-soluble?

- (a) Vitamin A
- (b) Vitamin B

(c) Vitamin D

(d) Vitamin E

Chhattisgarh P.C.S. (Pre) 2011

Ans. (b)

See the explanation of above question.

18. Some vitamins are fat soluble while others are water soluble. Which of the following are water soluble :

- (a) Vitamin A, B and D
- (b) Vitamin A, D, E and K
- (c) Vitamin A, B and C
- (d) Vitamin B and C

U.P. Lower Sub. (Pre) 2002

Ans. (d)

See the explanation of above question.

19. Which of the following is a micro-mineral found in human body fluids/tissues?

- (a) Phosphorus
- (b) Sulphur
- (c) Iodine
- (d) Magnesium

Rajasthan P.C.S. (Pre) 2024

Ans. (c)

Micro-minerals are those minerals, which comprise less than 0.01% of the total body weight and are required in concentrations of one part per million or less. They are also known as trace elements. Micro-minerals are essential nutrients needed in very small amounts by the body for various functions. Iron, zinc, copper, selenium, chromium, manganese, iodine, molybdenum, nickel, cobalt, fluorine etc. are important micro-minerals. Phosphorus, sulphur, magnesium, potassium, calcium, sodium and chlorine are macro-minerals, which are required in larger amounts in the body.

Hence, among the given options, option (c) is the correct answer.

20. Living organisms require at least 27 elements, of which, 15 are metals. Among these, those required in major quantities, include :

- (a) Potassium, manganese, molybdenum and calcium
- (b) Potassium, molybdenum, copper and calcium
- (c) Potassium, sodium, magnesium and calcium
- (d) Sodium, magnesium, copper and manganese

I.A.S. (Pre) 1996

Ans. (c)

Ample amount of potassium, sodium, magnesium, calcium, sulphur, phosphorus and chlorine are required for living organisms. They are known as macronutrients whereas micronutrients are needed in a small amount for the organisms.

21. Apples have special value for heart patients, because they are rich source of –

- (a) Sodium and Potassium
- (b) Phosphorus and Magnesium
- (c) Potassium and Phosphorus
- (d) Potassium only

U.P.P.C.S. (Mains) 2014

Ans. (d)

Potassium is an electrolyte mineral. It plays an important role in maintaining a healthy heart rhythm through muscle action. Apple trees absorb potassium in a greater amount than any other nutrient, so apples are a rich source of potassium. Apples are also rich sources of phytochemicals that can act as antioxidants.

22. Regular intake of fresh fruits and vegetables is recommended in the diet since they are a good source of antioxidants. How do antioxidants help a person maintain health and promote longevity?

- (a) They activate the enzymes necessary for vitamin synthesis in the body and help prevent vitamin deficiency.
- (b) They prevent excessive oxidation of carbohydrates, fats, and proteins in the body and help avoid unnecessary wastage of energy.
- (c) They neutralize the free radicals produced in the body during metabolism.
- (d) They activate certain genes in the cells of the body and help delay the aging process.

I.A.S. (Pre) 2011

Ans. (c)

An antioxidant is a molecule that inhibits the oxidation of other molecules. Oxidation is a chemical reaction involving the loss of electrons or an increase in oxidation state. Oxidation reactions can produce free radicals. In turn, these radicals can start chain reactions. When the chain reaction occurs in a cell, it can cause damage or death to the cell. Antioxidants terminate these chain reactions by removing free radical intermediates and inhibit other oxidation reactions. They do this by being oxidized themselves, so antioxidants are often reducing agents such as asthiols, ascorbic acid (vitamin C) or polyphenols. Fresh fruits and vegetables are a good source of antioxidants.

23. What is the role of anti-oxidants in human system?

- (a) They promote vitamin synthesis.
- (b) They prevent excessive oxidation of carbohydrates, protein and fat.

- (c) They protect cells against the effects of free-radicals.
(d) They activate certain genes which delay the ageing process.

U.P.P.C.S. (Mains) 2012

Ans. (c)

See the explanation of above question.

24. Among the following vegetables, the maximum Vitamin C is found in :

- (a) Chilli (b) Pumpkin
(c) Pea (d) Radish

U.P. P.C.S. (Mains) 2016

Ans. (a)

Among the above vegetables, the maximum vitamin C is found in chilli. Vitamin C present in each 100 gram of chilli, pumpkin, pea and radish is about 143.7, 9, 40 and 14.8 milligram respectively.

25. The vitamin which is found in citrus fruits and is necessary for maintaining healthy skin is –

- (a) Vitamin A (b) Vitamin B
(c) Vitamin C (d) Vitamin D

U.P.P.C.S. (Pre) 1999

Ans. (c)

Vitamin C occurs in good quantity in various citrus fruits like lemon, orange, Indian gooseberry (amla) etc. Vitamin C is a water-soluble vitamin which is necessary for normal growth and development and for maintaining healthy skin. Ascorbic acid is the chemical name of Vitamin C. Deficiency of this vitamin causes the disease scurvy in human.

26. The main source of Vitamin 'C' is –

- (a) Raw and fresh fruits (b) Milk
(c) Ghee (d) Pulses

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (a)

Vitamin A is the major constituent of milk and ghee which are enriched by fats while pulses are the rich source of protein. Raw and citrus fruits are major source of vitamin C.

27. The best source of Vitamin C is :

- (a) Apple (b) Mango
(c) Gooseberry (d) Milk

U.P.P.C.S. (Pre) 1995

Ans. (c)

Indian gooseberry (amla) is the richest source of vitamin C (Ascorbic acid). Orange, tomato, lemon, papaya, peas, etc. are also good sources of vitamin C. It is an important antioxidant which helps to protect against cancer, heart disease, stress, maintaining a healthy immune system and wound healing process. It is essential for sperm production and formation of collagen. The collagen protein involved in the building and health of cartilage, joints, skin and blood vessels etc.

28. Which of the following is the richest source of Vitamin C?

- (a) Orange (b) Apple
(c) Carrot (d) Amla

U.P. P.C.S.(Pre) 2018

Ans. (d)

See the explanation of above question.

29. Which of the following has the highest content of Vitamin C ?

- (a) Carrot (b) Guava
(c) Mango (d) Orange

Uttarakhand P.C.S. (Pre) 2002

Ans. (d)

See the explanation of above question.

30. The chemical name of Vitamin C is –

- (a) Ascorbic Acid (b) Thiamine
(c) Citric Acid (d) Tartaric acid

42nd B.P.S.C. (Pre) 1997

Ans. (a)

See the explanation of above question.

31. The richest source of Vitamin 'C' is –

- (a) Potato (b) Sugarcane
(c) Orange (d) Sugarbeet

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (c)

See the explanation of above question.

32. Which of the following vitamins is helpful for healing of a wound?

- (a) A (b) B
(c) C (d) D

Uttarakhand P.C.S. (Pre) 2006

U.P. P.C.S. (Mains) 2017

Ans. (c)

See the explanation of above question.

33. Vitamin 'C' help in absorption of which of following elements?

- (a) Iron (b) Calcium
(c) Iodine (d) Sodium

U.P.U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (a)

Vitamin C is helpful in iron absorption whereas vitamin D helps in absorption of magnesium and calcium.

34. Which of the following vitamins does not get stored in the body?

- (a) Vitamin A (b) Vitamin C
(c) Vitamin D (d) Vitamin E

U.P.R.O./A.R.O. (Mains) 2014

Ans. (b)

The body cannot store vitamin C. Vitamins A, D, E and K are fat-soluble, which means they bond with fat molecules and can be stored in the body's organs (particularly the liver), tissues or blood for long periods of time. Vitamin C, on the other hand is actually water-soluble and bonds instead of water, which is flushed out of the body via the urine rather quickly.

35. Which of the following vitamin's deficiency causes disease of slow blood coagulation ?

- (a) Vitamin C (b) Vitamin D
(c) Vitamin E (d) Vitamin K

R.A.S./R.T.S. (Pre) 1992

Ans. (d)

Vitamin K is a responsible factor for blood clotting or coagulation. Without vitamin K, blood coagulation is seriously impaired and uncontrolled bleeding occurs. Vitamin K is known as antihemorrhagic factor.

36. Which of the following vitamin helps in blood clotting in humans?

- (a) Vitamin A (b) Vitamin B
(c) Vitamin K (d) Vitamin D

M.P. P.C.S. (Pre) 2022

Ans. (c)

See the explanation of above question.

37. Which of the following vitamins is mainly responsible for blood coagulation in human being?

- (a) Vitamin K (b) Vitamin E

(c) Vitamin A

(d) Vitamin C

U.P.P.C.S. (Pre) 2020

Ans. (a)

See the explanation of above question.

38. Which one of the following vitamins is responsible for blood clotting ?

- (a) C (b) K
(c) E (d) D

U.P.P.C.S. (Mains) 2012

Uttarakhand P.C.S. (Pre) 2012

U.P.P.C.S. (Mains) 1991

Ans. (b)

See the explanation of above question.

39. Which of the following vitamin is effective in blood clotting?

- (a) Vitamin K (b) Vitamin A
(c) Vitamin B (d) Vitamin C

U.P.P.C.S. (Mains) 2008

Ans. (a)

See the explanation of above question.

40. The vitamin which is effective in blood clotting is :

- (a) Vitamin A (b) Vitamin B
(c) Vitamin D (d) Vitamin K
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

See the explanation of above question.

41. Which of the following vitamins is used as an antidote to anticoagulant poisons?

- (a) Vitamin A (b) Vitamin D
(c) Vitamin E (d) Vitamin K
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Vitamin 'K' is used as an antidote to anticoagulant poisons. This vitamin is called antihemorrhagic factor. Operation of persons with deficiency of vitamin K can not be done easily because there is a fear of excess bleeding.

42. Which of the following is synthesized by intestinal bacteria?

- (a) Vitamin B₁₂ (b) Vitamin C
(c) Vitamin K (d) Vitamin A

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (a) & (c)

Vitamin B₁₂ and vitamin K both are synthesized by bacteria in human intestine.

43. Which one of the following pairs is not correctly matched ?

- | | |
|---------------|-------------|
| (a) Niacin | – Pellagra |
| (b) Thiamin | – Beri-beri |
| (c) Vitamin D | – Rickets |
| (d) Vitamin K | – Sterility |

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (d)

The deficiency of Vitamin 'K' causes a problem in blood coagulation. Other pairs are correctly matched.

44. The source of vitamin D is –

- | | |
|------------|---------------|
| (a) Lemon | (b) Sun rays |
| (c) Orange | (d) Cashewnut |

U.P.P.C.S. (Pre) 1993

Ans. (b)

The source of vitamin D is sun rays. In fact, vitamin D is synthesized in our dermal cells by sun rays which is released in the blood. Besides of sun rays, vitamin D is also obtained from butter, the yolk of egg, liver, and kidney, etc. Rickets in children and osteomalacia in adults occur due to deficiency of vitamin D. Together with calcium, deficiency of vitamin D causes osteoporosis in older adults.

45. The vitamin available from sun rays is –

- | | |
|---------------|---------------|
| (a) Vitamin A | (b) Vitamin B |
| (c) Vitamin C | (d) Vitamin D |

Uttarakhand P.C.S. (Mains) 2006

M.P.P.C.S. (Pre) 1999

Ans. (d)

See the explanation of above question.

46. Which one of the following vitamins is formed in skin in the presence of sunlight?

- | | |
|---------------|---------------|
| (a) Vitamin A | (b) Vitamin D |
| (c) Vitamin K | (d) Vitamin E |

U.P. R.O./A.R.O. (Mains) 2017

Ans. (b)

See the explanation of above question.

47. The Deficiency of Vitamin D causes the disease :

- | | |
|------------------|---------------|
| (a) Rickets | (b) Beri-beri |
| (c) Osteoporosis | (d) Pellagra |

Uttarakhand P.C.S. (Mains) 2002

Ans. (a & c)

See the explanation of above question.

48. Which of the following is found in the composition of vitamin D?

- | | |
|-------------------|----------------|
| (a) Ascorbic acid | (b) Calciferol |
| (c) Folic acid | (d) Retinol |

Chhattisgarh P.C.S. (Pre) 2011

U.P.P.C.S. (Pre) 1997

Ans. (b)

There are two major chemical forms of vitamin D, namely vitamin D₂ sometimes referred to as ergocalciferol and vitamin D₃ sometimes referred to as cholecalciferol. Vitamin C is known as ascorbic acid whereas vitamin A is known as retinol. Folic acid is the name of Vitamin B₉.

49. Which of the following vitamins is known as ergocalciferol?

- | | |
|--|----------------------------|
| (a) Vitamin D ₂ | (b) Vitamin D ₃ |
| (c) Vitamin B ₁₂ | (d) Vitamin B ₆ |
| (e) None of the above/More than one of the above | |

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

See the explanation of above question.

50. The common name of the chemical compound 'Cholecalciferol' is :

- | | |
|------------------|---------------|
| (a) Bone Calcium | (b) Vitamin D |
| (c) Vitamin B | (d) Vitamin C |

U.P.P.C.S. (Pre) 2015

Ans. (b)

See the explanation of above question.

51. Which of the following vitamin is considered as hormone?

- | | |
|-------|-------|
| (a) A | (b) B |
| (c) C | (d) D |

I.A.S. (Pre) 1994

Ans. (d)

Vitamin D is considered as hormone. It is a fat-soluble vitamin which exists in various forms such as vitamin D₃ (cholecalciferol) and D₂ (ergocalciferol). Both D₂ and D₃ are not biologically active; they must be modified in the body to have any effect. The active form of vitamin D is known as 1,25-dihydroxyvitamin D [1,25 (OH)₂D] or calcitriol.

52. Deficiency of which vitamin can cause night blindness?

- | | |
|--------------------|-------|
| (a) B ₁ | (b) C |
| (c) A | (d) E |

Uttarakhand P.C.S. (Pre) 2010

Uttarakhand U.D.A./L.D.A. (Pre) 2007

U.P.P.C.S. (Mains) 2006

R.A.S./R.T.S. (Pre) 2003

M.P.P.C.S. (Pre) 1990

Ans. (c)

Vitamin A is needed by the retina of the eye in the form of retinol, which combines with protein opsin to form rhodopsin, the light absorbing molecules necessary for both low light (scotopic vision) and colour vision.

53. In human body, vitamin A is stored in –

- (a) Liver (b) Pancreas
(c) Spleen (d) Stomach

U.P.P.C.S. (Pre) 1997

Ans. (a)

Vitamin A is largely stored in the liver in the human body. Its chemical name is retinol whose deficiency causes night blindness.

54. In human body, Vitamin A is stored in –

- (a) Liver (b) Skin
(c) Lung (d) Kidney
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

U.P. Lower Sub. (Pre) 2015

Ans. (a)

See the explanation of above question.

55. Which one of the following contains more vitamin A content ?

- (a) Cabbage (b) Carrot
(c) Cauliflower (d) Spinach

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (b)

Carrot is the richest source of vitamin A. Spinach is a major source of iron which contributes to the level of haemoglobin concentration, beneficial for oxygen transportation.

56. Which of the following is the richest source of vitamin A?

- (a) Apple (b) Papaya
(c) Guava (d) Mango

U.P. Lower Sub. (Pre) 2015

Ans. (d)

Mango is a rich source of vitamin A among all the given fruits. It is also rich in vitamin C and E.

57. Which one of the following is not formed in our body?

- (a) Vitamin A (b) Protein
(c) Enzyme (d) Hormone

U.P. Lower Sub. (Mains) 2013

U.P.P.C.S. (Pre) 2011

Ans. (a)

Vitamin A cannot be synthesized by the human body.

58. Which of the following vitamins is essential for normal vision?

- (a) Folic acid (b) Riboflavin
(c) Niacin (d) Retinol

U.P. P.C.S. (Pre) 2021

Ans. (d)

Vitamin A is the name of a group of fat-soluble retinoids, including retinol, retinal, and retinyl esters. Vitamin A is involved in immune function, vision, reproduction, and cellular communication. Vitamin A is critical for vision as an essential component of rhodopsin, a protein that absorbs light in the retinal receptors, and because it supports the normal differentiation and functioning of the conjunctival membranes and cornea. Vitamin A deficiency may lead to 'night blindness'. Vitamin A also supports cell growth and differentiation, playing a critical role in the normal formation and maintenance of the heart, lungs, kidneys, and other organs.

59. Assertion (A) : If somebody stops taking green vegetables he will suffer from night blindness.

Reason (R) : He will suffer from vitamin A deficiency. Select the correct answer from the codes given below:

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2011

Ans. (d)

Vitamin A in humans is primarily obtained by conversion of beta-carotene, a pigment found in fruits and vegetables. Green vegetables are a good source for beta-carotene. If somebody takes green vegetables regularly, he gains sufficient amount of vitamin A. If he stops taking green vegetables, he could suffer from vitamin A deficiency, that may lead to the disease called 'night blindness'. However, vitamin A is found in many other foods also, so the person devoid of green vegetables does not necessarily suffer from night blindness.

60. Which of the following vitamins contain cobalt –

- (a) Vitamin B₆ (b) Vitamin B₂
(c) Vitamin B₁ (d) Vitamin B₁₂

48th to 52nd B.P.S.C. (Pre) 2008

U.P.P.C.S. (Pre) 2001

U.P.P.C.S. (Spl.) (Mains) 2004

U.P.P.C.S. (Mains) 2006

Ans. (d)

Vitamin B₁₂ (Cobalamin) is a water-soluble vitamin. It contains a metallic ion cobalt. Its chemical formula is C₆₃H₈₈CoN₁₄O₁₄P. It functions as a cofactor for enzymes in the metabolism of amino acids and fatty acids, required for new cell synthesis, normal blood formation and neurological function. Its deficiency causes pernicious anaemia, nervous system abnormality etc.

61. Which one of the following metals is present in the vitamin B₁₂?

- (a) Cobalt (b) Iron
(c) Zinc (d) Magnesium

U.P. P.C.S. (Mains) 2016

Ans. (a)

See the explanation of above question.

62. Which of the following vitamins contains cobalt?

- (a) Vitamin K (b) Vitamin B₁₂
(c) Vitamin B₆ (d) Vitamin B₂

U.P.R.O./A.R.O. (Mains) 2014

Ans. (b)

See the explanation of above question.

63. Consider the statements :

Assertion (A) : Vitamin B₁₂ remains stored in liver for about 3-5 years.

Reason (R) : Vitamin B₁₂ is water-soluble.

Select the correct answer using the codes given below:

Codes :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P. P.C.S. (Mains) 2017

Ans. (b)

Vitamin B₁₂ is also known as cobalamin which is water-soluble and remains stored in the liver for about 3-5 years. Vitamin B₁₂ is a co-factor of DNA synthesis and in both fatty acid and amino acid metabolism. It is particularly important for the normal functioning of the nervous system via its role in the synthesis of myelin and in the maturation of developing RBC in the bone marrow. It contains cobalt (Co).

64. Cyanocobalamin is –

- (a) Vitamin C (b) Vitamin B₂
(c) Vitamin B₆ (d) Vitamin B₁₂

U.P.P.C.S. (Pre) 1996

Ans. (d)

Cyanocobalamin is a manufactured form of vitamin B₁₂ which is used to treat vitamin B₁₂ deficiency.

65. Thiamine is :

- (a) Vitamin C (b) Vitamin B₂
(c) Vitamin B₆ (d) Vitamin B₁

U.P.P.C.S. (Pre) 1999

Ans. (d)

Thiamin or Thiamine (Vitamin B₁) is a water-soluble vitamin that is necessary for carbohydrate and amino acid metabolism. Its main sources are rice bran, whole wheat flour, eggs, yeast, meat, etc. Thiamine deficiency results in beri-beri, a disease characterized by multiple neuritides, general debility and heart failure.

66. A person will suffer from beri-beri, rickets and scurvy if he is not taking adequate amounts of :

- (a) Vitamins B₁₂, A and C (b) Vitamins B₁, D and C
(c) Vitamins A, B and E (d) Vitamins B₆, A and K

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

Diseases related with deficient intake of vitamins are as follows :

- Vitamin B₁ (Thiamin) – Beri-beri, Wernicke-Korsakoff syndrome
- Vitamin D (Ergocalciferol, Cholecalciferol) – Rickets and Osteomalacia
- Vitamin C (Ascorbic acid) – Scurvy

67. Which one of the following vitamins is helpful in keeping a dream in memory for a longer duration?

- (a) Vitamin A (b) Vitamin D
(c) Vitamin B₆ (d) Vitamin C

U.P. Lower Sub. (Pre) 2013

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (c)

Taking vitamin B₆, vitamin B₁₂ or folate (Vitamin B₉) may have an impact on the quality of a person's dreams and his ability to recall them.

68. Which of the following pairs is not correctly matched?

- (a) Vitamin B₁ – Oranges
(b) Vitamin D – Cod-liver oil
(c) Vitamin E – Wheat germ oil
(d) Vitamin K – Alfalfa
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Sunlight and cod-liver oil are rich source of vitamin D. Wheat germ oil is rich in vitamin E while alfalfa is rich in vitamin K. Though orange is rich in vitamin C but it is also a low source of vitamin B₁. Thus option (e) is the most probable answer.

69. Which of the following pairs is NOT correctly matched?

- | (Vitamin) | | (Rich Source) |
|----------------------------|---|----------------|
| (a) Vitamin B ₆ | – | Rice bran |
| (b) Vitamin B ₂ | – | Cod-liver oil |
| (c) Vitamin E | – | Wheat germ oil |
| (d) Vitamin K | – | Alfalfa |

U.P.P.C.S. (Pre) 2019

Ans. (b)

Cod-liver oil is a dietary supplement derived from liver of cod fish. It contains omega-3 fatty acids, vitamin A and vitamin D. Vitamin B₂, also known as riboflavin, is a vitamin found in eggs, green vegetables, milk and other dairy products, meat, mushrooms, and almonds. Vitamins and their rich sources given in other options are correctly matched.

70. Which one of the following pairs is NOT correctly matched?

- | | | |
|----------------------------|---|---------------|
| (a) Vitamin A | – | Orange |
| (b) Vitamin C | – | Lemon |
| (c) Vitamin D | – | Cod liver oil |
| (d) Vitamin B ₆ | – | Rice bran |

U.P. P.C.S. (Pre) 2021

Ans. (a)

Orange is a good source of several vitamins and minerals, especially vitamin C, thiamin, folate, calcium and potassium. The main sources of vitamin A include carrots, milk, butter, egg, cod liver oil etc. Other pairs are correctly matched.

71. Match List-I with List-II and select the correct answer using the code given below :

- | List-I | List-II |
|--------------|-------------------|
| A. Vitamin K | 1. Wheat germ oil |
| B. Vitamin D | 2. Lemon |
| C. Vitamin E | 3. Cod-liver oil |
| D. Vitamin C | 4. Alfa-alfa |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 1 | 4 | 2 |
| (c) | 2 | 3 | 4 | 1 |
| (d) | 4 | 3 | 1 | 2 |

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (d)

The correctly matched lists are as follows :

- | List-I (Vitamin) | | List-II (Source) |
|------------------|---|------------------|
| Vitamin K | – | Alfa-alfa |
| Vitamin D | – | Cod-liver oil |
| Vitamin E | – | Wheat germ oil |
| Vitamin C | – | Lemon |

72. Match List-I with List-II and choose the correct answer using the codes given below the lists :

- | List-I
(Vitamin) | List-II
(Disease) |
|---------------------|----------------------|
| A. Vitamin A | 1. Beriberi |
| B. Vitamin B | 2. Blindness |
| C. Vitamin C | 3. Rickets |
| D. Vitamin D | 4. Scurvy |

Codes :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 3 | 2 | 1 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 1 | 2 | 3 | 4 |
| (d) | 1 | 3 | 2 | 4 |

U.P.P.C.S. (Pre) 2024

Ans. (b)

The correctly matched lists are as follows :

- | Vitamin | | Disease |
|-----------|---|-----------|
| Vitamin A | – | Blindness |
| Vitamin B | – | Beriberi |
| Vitamin C | – | Scurvy |
| Vitamin D | – | Rickets |

Note : Night blindness is one of the first signs of Vitamin A deficiency. In its more severe forms, Vitamin A deficiency can lead to blindness, specifically through a condition called xerophthalmia, which causes the cornea to dry and become damaged, potentially leading to permanent vision loss.

73. Match List-I with List-II and select the correct answer from the code given below the lists :

- | List-I
(Disease) | List-II
(Cause) |
|---------------------|--------------------|
| A. Nightblindness | 1. Vitamin D |
| B. Rickets | 2. Vitamin C |
| C. Scurvy | 3. Vitamin B |
| D. Beri-beri | 4. Vitamin A |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 4 | 1 | 2 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 2 | 3 | 4 | 1 |

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (c)

Nightblindness is caused due to deficiency of vitamin A, Rickets caused due to deficiency of vitamin D, Scurvy is caused by deficiency of vitamin C and Beri-beri is the result of deficiency of vitamin B₁.

74. Match List-I with List- II and select the correct answer from the code given below the Lists :

List- I (Disease)	List- II (Cause)
A. Rickets	1. Vitamin D
B. Beri-beri	2. Vitamin B ₁
C. Nyctalopia	3. Vitamin A
D. Scurvy	4. Vitamin C

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	2	3	1
(c)	3	2	1	4
(d)	None of the above			

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

Rickets is caused by deficiency of vitamin D and Beri-beri is caused by deficiency of vitamin B₁. Nyctalopia or night blindness is caused due to deficiency of vitamin A and scurvy is caused due to deficiency of vitamin C.

75. Match List-I with List - II and select the correct answer using the code given below the lists :

List - I	List-II
A. Vitamin C	1. Night blindness
B. Folic acid	2. Beri-beri
C. Vitamin A	3. Anaemia
D. Vitamin B ₁	4. Scurvy

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	2	3	1	4
(c)	4	3	2	1
(d)	1	2	4	3

U.P.P.C.S. (Pre) 2015

U.P. R.O./A.R.O. (Pre) 2023

Ans. (a)

The correctly matched order is –
 Vitamin C – Scurvy
 Folic acid – Anaemia
 Vitamin A – Night blindness
 Vitamin B₁ – Beri-beri

76. Which one of the following pairs is no correctly matched?

(a) Vitamin A	–	Night Blindness
(b) Vitamin B ₃	–	Pellagra
(c) Vitamin D	–	Color Blindness
(d) Folic acid	–	Anaemia

U.P. R.O./A.R.O. (Pre) 2016

Ans. (c)

Color blindness is a sex-linked genetic disorder. It does not relate to vitamin D. The deficiency of vitamin D causes rickets and osteomalacia. The deficiency of vitamin A causes night blindness. Vitamin B₃ is also called niacin. Its deficiency causes pellagra. The deficiency of folic acid (vitamin B₉) causes anaemia.

77. Consider the following pairs :

Vitamin	Deficiency disease
1. Vitamin C	: Scurvy
2. Vitamin D	: Rickets
3. Vitamin E	: Night blindness

Which of the pairs given above is/are correctly matched?

(a) 1 and 2 only	(b) 3 only
(c) 1, 2 and 3	(d) None

I.A.S. (Pre) 2014

Ans. (a)

Scurvy is a disease resulting from the deficiency of vitamin C. Rickets is a nutritional disorder that can develop due to deficiency of vitamin D while vitamin A deficiency can lead to night blindness. So it is clear that only pairs 1 and 2 are correctly matched.

78. Riboflavin deficiency causes which of the following diseases?

(a) Pellagra	(b) Scurvy
(c) Beri Beri	(d) Cheilosis

U.P. P.C.S. (Pre) 2021

Ans. (d)

The riboflavin (Vitamin B₂) deficiency (also known as ariboflavinosis) causes skin disorders, hyperemia (excess blood) and edema of the mouth and throat, angular stomatitis (lesions at the corners of the mouth), cheilosis (swollen, cracked lips), hair loss, reproductive problems, sore throat, itchy and red eyes, and degeneration of the liver and nervous system. People with riboflavin deficiency typically have deficiencies of other nutrients, so some of these signs and symptoms might reflect these other deficiencies.

79. Which one of the following vitamins and deficiency diseases is wrongly matched ?

(a) Calciferol	–	Osteomalacia
(b) Niacin	–	Pellagra

- (c) Cobalamin – Pernicious anaemia
(d) Riboflavin – Beri-beri

U.P.P.C.S. (Mains) 2002

Ans. (d)

Beri-beri disease is caused by a deficiency of Vitamin B₁ whose chemical name is Thiamin. Other options are correctly matched. Riboflavin is the chemical name of Vitamin B₂ which is responsible for stomatitis, cheilosis and malnutrition.

80. Which one of the following pairs is not correctly matched?

- (a) Vitamin D – Rickets
(b) Vitamin C – Bleeding gums
(c) Vitamin A – Rheumatism
(d) Vitamin B₁ – Beriberi

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

Lack of vitamin A causes night blindness, not rheumatism. Other pairs are correctly matched.

81. Match List- I with List- II and select the correct answer from the code given below the lists :

List I	List II
A. Retinol	1. Pernicious anaemia
B. Tocopherol	2. Mental illness
C. Cyanocobalamin	3. Conjunctivitis
D. Pyridoxine	4. Sterility/infertility

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	1	2
(c)	4	3	2	1
(d)	3	4	2	1

U.P. Lower Sub. (Pre) 2002

U.P. Lower Sub. (Spl.) (Pre) 1999

Ans. (b)

Retinol (vitamin A)	–	Conjunctivitis
Tocopherol (vitamin E)	–	Sterility
Cyanocobalamin (vitamin B ₁₂)	–	Pernicious anaemia
Pyridoxine (vitamin B ₆)	–	Mental illness

82. Which of the following pairs is not correctly matched?

- (a) Retinol - Xerophthalmia
(b) Tocopherol - Beri-beri
(c) Cyanocobalamin - Anaemia
(d) Ergocalciferol - Rickets
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Deficiency of retinol i.e. vitamin A, leads to Xerophthalmia. Lack of cyanocobalamin (vitamin B₁₂) causes pernicious anaemia and lack of vitamin D (Cholecalciferol 'D₃' and Ergocalciferol 'D₂') causes rickets in children. Beri-beri is caused by lack of vitamin 'B₁' i.e. thiamin not due to lack of tocopherol (vitamin 'E').

83. Which of the following pairs is not correctly matched?

Vitamin	Role
(a) Vitamin B ₁₂	– antianaemic factor
(b) Vitamin C	– antiscorbutic factor
(c) Vitamin D	– antisterility factor
(d) Vitamin K	– antihaemorrhagic factor

U.P. P.C.S. (Pre) 2017

Ans. (c)

Vitamin E (Tocopherol) works as antisterility factor. Lack of vitamin D causes rickets in children and osteomalacia in adults. Pairs of other options are correctly matched.

84. Which among the following pairs is not correctly matched?

- (a) Thiamin – Beri-beri
(b) Vitamin D – Rickets
(c) Vitamin K – Sterility
(d) Niacin – Pellagra

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (c)

Napthoquinone is the chemical name of vitamin K. It is known as anti-hemorrhagic factor. Its sources are green leaves, egg, liver & intestinal bacteria etc. Sterility is related with deficiency of vitamin E. Other pairs are correctly matched.

85. Which one of the following is an important source of vitamin E?

- (a) Palm oil (b) Coconut oil
(c) Wheat germ oil (d) Mustard oil

U.P. P.C.S. (Pre) 2016

Ans. (c)

The Wheat germ oil is extracted from the germ of the wheat kernel which makes up only about 2.5% by weight of the kernel. Wheat germ oil has the highest content of vitamin E. It is a fat-soluble vitamin and plays a role as an antioxidant in the body.

86. Which one of the following compounds is not a vitamin?

- (a) Niacin (b) Thyroxine
(c) Riboflavin (d) Pyridoxine

U.P. P.C.S. (Mains) 2017

Ans. (b)

Niacin - Vitamin B₃, Riboflavin - Vitamin B₂ and Pyridoxine - Vitamin B₆ are vitamins while Thyroxine is a hormone secreted by the thyroid gland (biggest endocrine gland of a human) located in the neck.

87. Banana accepted as most calories fruit has what amount of calories per 100 gm –

- (a) 416 Kcal (b) 316 Kcal
(c) 216 Kcal (d) 116 Kcal

U.P. Lower Sub. (Pre) 1998

Ans. (d)

Banana is the berry fruit of family 'Musaceae' and its botanical name is *Musa balbisiana*. It has a large amount of carbohydrates. It also contains fat, proteins, mineral and vitamins. About 89 Kcal/100 gm energy is obtained from banana. In the given options, option (d) is near about 89 Kcal.

88. Legumes are highly nutritious because they are rich in :

- (a) Fat (b) Protein
(c) Oil (d) Starch
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Legumes are highly nutritious because they are rich in proteins. Legumes are typically low in fat, contain no cholesterol, and are high in folate, potassium, iron and magnesium. As a good source of protein, legumes can be a healthy substitute for meat, which has more fat and cholesterol.

89. The highest protein content is found in –

- (a) Pigeon pea (b) Soyabean
(c) Horse pea (d) Wheat

**U.P.P.C.S. (Spl.) (Mains) 2004
Uttarakhand P.C.S. (Pre) 2002**

Ans. (b)

Soyabean is the richest source of plant protein. It contains about 40% protein. It is classified as an oilseed as well as pulse crop. The milk is produced by soyabean which is equivalent to the milk of a cow. Each grain of soyabean contains about 40% protein, 20-23% fat and 30% carbohydrate.

90. Which of the food grains is richest in protein?

- (a) Gram (b) Pea
(c) Soyabean (d) Wheat

**U.P.P.C.S. (Mains) 2008
40th B.P.S.C. (Pre) 1995**

Ans. (c)

See the explanation of above question.

91. The highest protein content is found in the grain of –

- (a) Black gram (b) Pigeon pea
(c) Field pea (d) Soyabean

U.P.R.O/A.R.O. (Pre) 2014

Ans. (d)

See the explanation of above question.

92. Which one of the following is the richest source of protein?

- (a) Rice (b) Groundnut
(c) Apple (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

Among the given options, groundnut is the richest source of protein. Rice is the best source of carbohydrate and apple is known for the richness of antioxidants.

93. A rich source of both protein and fat is :

- (a) Coconut (b) Groundnut
(c) Soyabean (d) Sunflower

U.P.P.C.S. (Mains) 2008

Ans. (b)

The groundnut is a rich source of both proteins and fat. Groundnut contains about 25.3% protein and 48.1% fat. In groundnut amount of protein is about 1.3 times of meat, 2.5 times of eggs and 8 times of fruits.

94. Which of the following is mostly carbohydrate?

- (a) Wheat (b) Barley
(c) Rice (d) Maize
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Among the given options, rice has the highest percentage of carbohydrates.

95. Which one of the following statements is not correct regarding the amino acids, tyrosine, present in protein rich food?

- (a) It can help increase energy levels.
(b) It can help with emotional & environmental stress.
(c) It can combat depression.
(d) It can protect against free radicals responsible for aging.

U.P. P.C.S. (Pre) 2010

Ans. (d)

Tyrosine is an amino acid which helps in the formation of neurotransmitters (adrenaline, noradrenaline, dopamine) in the brain. Human feels laziness and woeful due to lack of tyrosine. Tyrosine helps in enhancement of physical vigilance and energy. But it cannot protect against free radicals responsible for aging.

96. The substance responsible for bread-making quality in wheat is –

- (a) Gluten (b) Globulin
(c) Glycin (d) Lycine

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (a)

Gluten is the protein composite, found in wheat. Gluten gives elasticity to dough, helping its growth and keep its shape and often gives the final product a chewy texture.

97. Daily energy required by a hard worker man is-

- (a) 3000 kilo calorie (b) 2700 kilo calorie
(c) 4000 kilo calorie (d) 6000 kilo calorie

R.A.S./R.T.S.(Pre) 2012

Ans. (c)

The total energy requirement for a man during very hard work under different conditions is about 3500-4000 kilo calories.

98. How many calories are gained by drinking a glass of water?

- (a) Zero (b) 15
(c) 25 (d) 50

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

Water does not contain fat, carbohydrates or proteins. Therefore, it does not contain any calories.

99. An essential constituent of diet is -

- (a) Starch (b) Glucose
(c) Carbohydrate (d) Cellulose

U.P.R.O./A.R.O. (Pre) 2014

Ans. (c)

Carbohydrate is an essential constituent of diet. Carbohydrates are compounds of carbon, hydrogen and oxygen with a ratio of two hydrogens for every oxygen atom.

100. The major food fuel during exercise of long duration is

- (a) Fat (b) Protein
(c) Carbohydrate (d) None of these

Chhattisgarh P.C.S. (Pre) 2011

Ans. (c)

The major food fuel during exercise of high intensity and long duration is carbohydrate.

101. Nitrogen is NOT an essential constituent of :

- (a) Chlorophyll (b) RNA
(c) DNA (d) Carbohydrate

U.P. P.C.S. (Pre) 2021

Ans. (d)

Nitrogen is an essential constituent of DNA, RNA (and all amino acids/proteins) and chlorophyll, while it is not an essential component of carbohydrates. A carbohydrate is a biomolecule consisting essentially of carbon (C), hydrogen (H) and oxygen (O) atoms, usually with a hydrogen-oxygen atom ratio of 2 : 1 and thus with empirical formula $C_x(H_2O)_y$. Although all carbohydrates contain carbon, hydrogen, and oxygen, there are some that also contain nitrogen, phosphorus and sulphur.

102. The element that maintains balance between acidity and alkalinity in our body is

- (a) Phosphorus (b) Sodium
(c) Potassium (d) Calcium

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (a)

Phosphorus (in the form of phosphate) is the element that maintains a balance between acidity and alkalinity in our body.

103. Which one of the following compounds is not stored in human body –

- (a) Glycogen (b) Amino acids
(c) Fat (d) Tocopherol

U.P.P.C.S. (Mains) 2015

Ans. (b)

Unlike some macronutrients, including fats and starches, amino acids are not stored in human body and must be replenished daily through dietary intake. Proteins are polymers of amino acids.

104. Among the following which fruit is a rich source of iron?

- (a) Jambo (b) Karonda
(c) Loquat (d) Guava

U.P.U.D.A./L.D.A. (Pre) 2006

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

The fruit karonda is a rich source of iron, so it is sometimes used in the treatment of anaemia. It contains a fair amount of vitamin C which helps in healing wounded therefore is an antiscorbutic.

105. Consider the following minerals :

- 1. Calcium 2. Iron 3. Sodium**

Which of the minerals given above is/are required by human body for the contraction of muscles?

- (a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2013

Ans. (d)

Your muscles contract every time you move and the process begins when your muscles receive the signals from your brain. Thus, your muscles shorten or contract as different muscles fibres sides along each other. For proper muscles function, you not only need to get enough protein to build muscles and calories to fuel their contraction but also the right minerals like calcium, iron, sodium, magnesium and potassium to allow contraction to occur.

106. Spinach leaves contains the highest amount of –

- (a) Vitamin (b) Iron
(c) Fat (d) Carbohydrate

M.P.P.C.S. (Pre) 1993

Ans. (b)

Spinach has a high nutritional value for the human body. It is a rich source of iron and calcium. According to the United States Department of Agriculture, 180 gram serving of boiled spinach contains about 6.43 mg of iron, whereas one 170 gram ground hamburger patty contains almost 4.42 mg of iron. Thus, spinach contains a relatively high level of iron, compared to other vegetables and meat sources.

107. Which one of the following is a rich source of iron?

- (a) Carrot (b) Pea
(c) Rice (d) Spinach

U.P.P.C.S. (Pre) 2008

Ans. (d)

See the explanation of above question.

108. Iron content is found maximum in :

- (a) Beans (b) Eggs
(c) Green vegetables (d) Milk

Uttarakhand P.C.S. (Pre) 2002

Ans. (c)

Green vegetables are the richest source of iron. Other high iron foods are liver, sunflower seeds, nuts, beef, lamb, beans, whole grains, dark leafy greens (spinach), dark chocolate and tofu. Iron present in haemoglobin and cytochrome enzyme is an essential mineral used to transport oxygen to all parts of the body. A deficiency in iron causes anaemia and a chronic deficiency can lead to organ failure.

109. Grains of basmati rice get elongated on being cooked because of higher content of –

- (a) Lysine (b) Amylose
(c) Sugar (d) Oil

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

There are many different varieties of rice. They differ in amounts of nutrition and more importantly the type of starch. There are two types of starch in rice : amylose and amylopectin. Amylose is a long, straight starch molecule that does not gelatinize during cooking. So, rice which contains more of this starch tends to cook fluffy, with separate grains. Long grain white rice (basmati) has the most amylose and the least amylopectin, so it tends to be the fluffiest and least sticky. Medium grain rice has more amylopectin and short grain rice has even more amylopectin with least or no amylose.

110. Among the following, presence of which brings sweetness in milk?

- (a) Microse (b) Lactose
(c) Sucrose (d) Carotene

U.P.P.C.S. (Pre) 2007

Ans. (b)

Lactose is a disaccharide sugar derived from glucose and galactose. It is responsible for imparting sweetness to milk.

111. The saccharide present in milk is –

- (a) Sucrose (b) Lactose
(c) Maltose (d) Cellobiose

U.P.P.S.C. (R.I.) 2014

Ans. (b)

Three common disaccharides :

Sucrose – Common table sugar = glucose + fructose

Lactose – Major sugar in milk = glucose + galactose

Maltose – Product of starch digestion = glucose + glucose

112. Which bacteria cause milk to spoil ?

- (a) Aspergillus (b) Staphylococcus
(c) Pseudomonas (d) Lactobacillus

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Chhattisgarh P.C.S. (Pre) 2005

Ans. (d)

Lactobacillus bacteria causes milk to spoil. It is a very important bacteria which is found in the intestinal tract. Lactobacillus uses lactose for energy and creates lactic acid as a by-product. It is the lactic acid which makes milk taste sour. It is used to make dairy products like cheese, curd etc. Apart from this, it is also used to control cholesterol.

113. Which of the following converts milk into curd ?

- (a) Bacteria
- (b) Vitamin
- (c) Enzyme
- (d) None of the above

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (a)

See the explanation of above question.

114. Cow's milk is slightly yellow in colour due to the presence of –

- (a) Xanthophyll
- (b) Riboflavin
- (c) Vitamin B₁₂
- (d) Carotene

U.P.P.C.S. (Spl) (Mains) 2008

R.A.S./R.T.S.(Pre) 1999

U.P.P.C.S. (Pre) 1992

Ans. (d)

The yellow colour of cow's milk is due to the presence of carotene. It is a hydrocarbon whose formula is $C_{40}H_{x}$. It is found in hay, grass, green leaves, some fruits, carrots etc. When cow consumes carotene containing foods, some of the pigment is converted into vitamin A and some found unchanged in milk. Vitamin A imparts no colour to milk, whereas carotene gives milk a yellow colour.

115. Cow milk is yellowish white in colour due to the presence of –

- (a) Casein
- (b) Lactose
- (c) Carotene together with casein
- (d) Butyric acid together with lactose

U.P.R.O./A.R.O. (Pre) 2014

Ans. (c)

Cow milk is yellow in colour because of the presence of carotene. Buffalo milk appears whiter than cow milk because it does not contain the yellow pigment carotene. Casein is a protein that makes up approximately 80 percent of the total protein content of cow's milk. Existing as globules within the milk and largely responsible for giving milk its white colour.

116. What is the average fat content of buffalo milk ?

- (a) 7.2%
- (b) 4.5%
- (c) 9.0%
- (d) 10.0%

I.A.S. (Pre) 1997

Ans. (a)

The average fat content in buffalo milk is about 7 to 7.5%, while fat content in cow milk ranges from about 3.5 to 4.0 %. Apart from this it also contains minerals and other essential nutritions.

117. Rennin and lactase, the enzymes required to digest milk, disappear in the human body by the age of –

- (a) Two
- (b) Three
- (c) Five
- (d) Eight

I.A.S. (Pre) 1994

Ans. (*)

The enzymes necessary to break down and digest milk (rennin and lactase) start to disappear by the age of two or three years and completely disappear between 6 to 15 years of age in most humans. Therefore from around that age, many children no longer secrete those enzymes (lactase and renin) and they become what's referred to as 'lactose intolerant'. Many people around the world have known allergies to milk and milk products and are unable to digest such substances. In particular, many people are known to be unable to consume milk product due to such intolerance.

118. Milk protein is digested by which of the following enzyme-

- (a) Pepsin
- (b) Trypsin
- (c) Rennin
- (d) Erepsin

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (c)

Rennin enzymes play an important role in helping young mammals to digest milk. Rennin enzymes are produced by the stomach cell. It is secreted in large amounts right after the birth and then its production gradually drops off. When milk enters the stomach, prorenin is activated by the hydrochloric acid in the gastric juice and converted into the active rennin enzyme. The function of rennin enzyme is to curdle milk and separate it into semi-solid curds and liquid whey. Curdling of the milk is necessary to be retained in the stomach long enough for the milk proteins to be digested properly.

119. Casein is milk :

- (a) Bacterium
- (b) Sugar
- (c) Protein
- (d) Fat

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (c)

Casein is a phosphoprotein, which is found in milk. Phosphoprotein means any protein containing bound phosphate. The white colour of the milk is also due to the presence of casein. Some of the other proteins like lactoglobulin and lactalbumin are also found in milk.

120. White colour of milk is due to the presence of –

- (a) Lactose
- (b) Albumin
- (c) Carotene
- (d) Casein

U.P.P.C.S.(Pre) 2012

Jharkhand P.C.S. (Pre) 2016

Ans. (d)

See the explanation of above question.

121. Which of the following proteins is found in milk ?

- (a) Agglutinin (b) Casein
(c) Myosin (d) Haemoglobin

Uttarakhand P.C.S. (Pre) 2006

Ans. (b)

See the explanation of above question.

122. The tissues are formed in body by-

- (a) Fat (b) Carbohydrate
(c) Protein (d) Vitamins

U.P.P.C.S. (Pre) 1990

Ans. (c)

Proteins are large, complex molecules that play many critical roles in the body. They do most of the work in cells and are required for the structure, function and regulation of the body's tissues and organs. Serving as the basic structural molecule of all the tissues in the body, proteins makes up nearly 17 percent of the total body weight.

123. The slow growth of the human body is caused due to the deficiency of which of the following?

- (a) Fats (b) Vitamins
(c) Proteins (d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (c)

Proteins are an organic compound that contain mainly the element nitrogen, carbon, hydrogen, and oxygen. Proteins are the most diverse group of biologically important substances and are often considered to be the central compound, necessary for life. The body's primary building block for muscles, bone, skin and many other tissues is protein. So the slow growth of human body is caused due to the lack of protein.

124. Which of the following elements are present in all proteins?

- 1. Carbon 2. Hydrogen**
3. Oxygen 4. Nitrogen

Code :

- (a) 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 1995

Ans. (d)

Proteins, after water are the most abundant constituent of protoplasm. They are made up of carbon, oxygen, hydrogen and nitrogen but sulphur, iodine, iron, and some other elements may also be found in them in traces. Proteins are long chain compounds; their simple components or basic building units are the amino acids.

125. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	List-II
(molecules)	(elements present)
A. Vitamin B₁₂	1. Magnesium
B. Haemoglobin	2. Cobalt
C. Chlorophyll	3. Copper
D. Brass	4. Iron

Code :

	A	B	C	D
(a)	2	4	1	3
(b)	2	1	3	4
(c)	4	1	2	3
(d)	3	4	2	1

U.P.P.C.S. (Pre) 2014

U.P.P.C.S. (Pre) 2000

Ans. (a)

The Correctly matched list are as follows :

Vitamin B ₁₂	—	Cobalt
Haemoglobin	—	Iron
Chlorophyll	—	Magnesium
Brass	—	Copper

126. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	List-II
(molecules)	(elements present)
A. Vitamin B₁₂	1. Magnesium
B. Haemoglobin	2. Cobalt
C. Chlorophyll	3. Copper
D. Chalcopryrite	4. Iron

Code :

	A	B	C	D
(a)	2	4	1	3
(b)	2	1	3	4
(c)	4	1	2	3
(d)	3	4	2	1

U.P.P.C.S. (Mains) 2011

Ans. (a)

The Correctly matched list are as follows :

Vitamin B ₁₂	—	Cobalt
Haemoglobin	—	Iron
Chlorophyll	—	Magnesium
Chalcopryrite	—	Copper

127. Match List-I and List-II and find the correct answer from the codes given below :

List-I	List-II
A. Vitamin B ₁	1. Cobalamin
B. Vitamin B ₂	2. Calciferol
C. Vitamin B ₆	3. Riboflavin
D. Vitamin B ₁₂	4. Thymine
	5. Pyridoxine

Code :

A	B	C	D
(a) 2	3	5	4
(b) 4	3	5	1
(c) 1	2	3	4
(d) 4	3	2	1

U.P.P.S.C. (GIC) 2010

Ans. (b)

The list of vitamins and their names is as follows :

Vitamin A	—	Retinol
Vitamin B ₁	—	Thiamin
Vitamin B ₂	—	Riboflavin
Vitamin B ₃	—	Niacin
Vitamin B ₅	—	Pantothenic acid
Vitamin B ₆	—	Pyridoxine
Vitamin B ₇	—	Biotin
Vitamin B ₉	—	Folic acid
Vitamin B ₁₂	—	Cobalamin
Vitamin C	—	Ascorbic acid
Vitamin D	—	Calciferol
Vitamin E	—	Tocopherol
Vitamin K	—	Naphthoquinone

128. Which one of the following pairs is correctly matched ?

- (a) Vitamin B₁ — Retinol
 (b) Vitamin B₂ — Riboflavin
 (c) Vitamin B₆ — Niacin
 (d) Vitamin C — Calciferol

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (b)

See the explanation of above question.

129. Which one is not micronutrient?

- (a) Iron (b) Zinc
 (c) Sulphur (d) Manganese

U.P.R.O./A.R.O. (Mains) 2014

Ans. (c)

Micronutrients are nutrients, required by organisms throughout life in small quantities to regulate a range of physiological function. The micronutrients include iron, boron, copper, chloride, manganese, nickel, zinc and molybdenum while sulphur, carbon, hydrogen, oxygen, nitrogen, potassium, magnesium, calcium and phosphorus are macronutrients.

130. Which is the main vitamin found in Papaya?

- (a) Vitamin 'A' (b) Vitamin 'C'
 (c) Vitamin 'B' (d) Vitamin 'K'

M.P.P.C.S. (Pre) 1998

Ans. (b)

Christopher Columbus called papaya 'The fruit of the angel'. The main vitamin found in Papaya is vitamin C. One medium size (200 gm) papaya contains about 226 mg vitamin C - that is two to three times more than our recommended daily dose and it is also a good source of folate, fibre, vitamin A, magnesium, potassium, copper and pantothenic acid.

131. Which one of the following is not a natural source of Vitamin A ?

- (a) Mango (b) Papaya
 (c) Carrot (d) Milk

U.P.P.C.S. (Mains) 2009

Ans. (d)

Mango, papaya, and carrot are the natural sources of vitamin A, while milk is not a natural source of vitamin A. Milk contains very small amount of vitamin A.

132. Which one of the following does not denature a protein?

- (a) Heat
 (b) Infrared rays
 (c) X-rays
 (d) Heavy metal salts

U.P. P.C.S. (Mains) 2016

Ans. (b)

Heat, X-rays, heavy metal salts, etc. denature a protein whereas infrared rays do not.

133. Choose wrong statement regarding proteins :

- (a) We obtain proteins from both animals and plants.
 (b) Proteins are made of amino acids.
 (c) About 30 percent of our body weight comes from proteins.
 (d) Proteins are called building blocks of our body.

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

Proteins are called building blocks of our body. As per the NCERT, about 20 percent of our body weight comes from proteins. Proteins are essential for growth and repair of muscle and other body tissues. Proteins are made of amino acids that contain the elements carbon, hydrogen, oxygen and nitrogen. We obtain proteins from animal as well as vegetable sources. Proteins of animal origin are found in milk, eggs, cheese, fish and meat. Proteins from these sources contain all the essential amino acids (EAA) in adequate amounts. Vegetable proteins are found in pulses (legumes), cereals, beans, nuts, oilseeds, etc.

134. Protein requirement of young and old person are –

- (a) Almost the same
- (b) Higher in the young and lower in the old
- (c) Higher in the old and lower in the young
- (d) Very high in the young and very low in the old.

U.P. Lower Sub. (Mains) 2013

Ans. (c)

Elderly individuals are more likely to be influenced by various biological, environmental and social factors. This would result to increase protein needs for adults as compared to younger one. Thus in practice, the protein needs of elderly people are probably higher than that for the young.

135. Daily intake of proteins, recommended for a moderately active woman is :

- (a) 30 g
- (b) 37 g
- (c) 40 g
- (d) 45 g

Jharkhand P.C.S. (Pre) 2010

I.A.S. (Pre) 1997

Ans. (d)

We need protein for our muscles, bones and the rest of our body. Exactly how much protein we need with respect to our age are given below :

- Babies need about 10 grams per day.
- School-age kids need about 19-34 grams per day.
- Teenage boys need up to 52 grams per day.
- Teenage girls need about 46 grams per day.
- Adult men need about 56 grams per day.
- Adult women or moderately active woman need about 45-46 grams per day (65-70 grams, if pregnant or breast-feeding).

136. How much protein a lactating mother requires in a daily diet?

- (a) 30 grams
- (b) 20 grams
- (c) 70 grams
- (d) 100 grams

U.P.P.C.S. (Mains) 2008

Ans. (c)

See the explanation of above question.

137. Which one of the following statements is not correct ?

- (a) Milk contains none of the B vitamins.
- (b) Vitamin A (retinol) deficiency leads to the dry and scaly skin.
- (c) One of the symptoms of scurvy is pain in the joints.
- (d) Vitamin B₁ (thiamine) deficiency can lead to heart failure.

I.A.S. (Pre) 2003

Ans. (a)

Milk generally contains vitamins of the B vitamin group such as Thiamin (B₁), Riboflavin (B₂), Niacin (B₃), Cobalamin (B₁₂), Folate (B₉) and Pyridoxine (B₆). Thus statement of option (a) is incorrect while other given statements are correct.

138. Mother's milk provides almost complete diet to the new-born, but it lacks the following

- (a) Iron
- (b) Calcium
- (c) Magnesium
- (d) Potassium

U.P.P.C.S. (Pre) 2019

Ans. (a)

Mother milk provides almost complete diet to the new-born but it lacks iron nutrient. During the first few days after delivery, the mother produces colostrum. It is rich in protein and antibodies that provide passive immunity to the baby. Colostrum also helps the newborn's digestive system to grow and function properly. Mother's milk contains about 4.2% fat, 1.1% protein, 7.5% carbohydrates and 0.2% minerals. For better health of the new-born, the mother's milk contains zinc, calcium, vitamin A, B₆, B₁₂, potassium, magnesium, phosphorus etc. in adequate quantity.

139. Which of the following are characteristics shown in a breast-fed baby when compared to a bottlefed baby ?

1. It is less obese.
2. It shows, greater capacity to resist diseases.
3. It gets more vitamins and protein.
4. Its growth in height is abnormal.

- (a) 1, 2 and 3
- (b) 1, 2 and 4
- (c) 1, 3 and 4
- (d) 2, 3 and 4

I.A.S. (Pre) 1994

Ans. (a)

Health professionals consider mother's milk to be ideal nourishment for her baby. Mother's milk contains immunological protection against diseases. This happens because babies receive antibodies passed onto them from

their mother's milk. Breast milk contains more nutrients (vitamin and proteins). Breastfed babies are more healthy and less obese than the bottlefed babies. Abnormal growth in height is not a characteristic shown by a breastfed baby.

140. Besides proteins and carbohydrates, other elements of nutritional value found in milk, include :

- (a) Calcium, potassium, and iron
- (b) Calcium and potassium
- (c) Potassium and iron
- (d) Calcium and iron

I.A.S. (Pre) 1996

Ans. (b)

Milk contains about 87% water. It has proteins, vitamin A, vitamin B₁₂, B₁, B₃, B₉, B₆ and Riboflavin, carbohydrates, calcium (Ca) and potassium (K) etc. Thus, milk is an ideal diet containing almost all nutritious elements. Milk contains only a trace amount of iron.

141. Milk is a poor source of :

- (a) Calcium
- (b) Protein
- (c) Vitamin C
- (d) Carbohydrate
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Milk is considered a complete food because it provides protein, vitamin A, vitamin B₁, B₂, B₃, B₆, B₁₂, B₉, carbohydrate, calcium and potassium etc. Milk contains very small amounts of vitamin C. Hence, option (c) is the correct answer.

142. Starch is a :

- (a) Monosaccharide
- (b) Disaccharide
- (c) Polysaccharide
- (d) None of the above

M.P. P.C.S. (Pre) 2016

Ans. (c)

Starch is a polymeric carbohydrate consisting a large number of glucose units joined by glycoside bonds. This polysaccharide is produced by most green plants as an energy store.

143. Which one of the following statements regarding starch and cellulose is not correct ?

- (a) Both of them are of plant origin
- (b) Both of them are polymers
- (c) Both of them give colour with iodine
- (d) Both of them are made up of glucose molecules

I.A.S. (Pre) 1998

Ans. (c)

Starch and cellulose are two very similar polymers. In fact, they both are made from the same monomer, glucose (C₆H₁₂O₆) and have the same glucose-based repeat units. Starch gives blue-black colour in an iodine solution, whereas cellulose give no colour in this solution. Starch is an important component of stored food of plants, whereas cellulose is found in the cell wall of plants.

144. Which of the following would lead to malnutrition ?

- 1. Overnutrition
- 2. Undernutrition
- 3. Imbalanced nutrition

Select the correct answer by using the codes given below :

Code :

- (a) 2 alone
- (b) 2 and 3
- (c) 1 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 1996

Ans. (d)

Malnutrition is a condition that results from eating a diet in which nutrients are either not enough or are too much such that it causes health problems. Imbalanced nutrition includes both undernutrition and overnutrition. Thus, option (d) is the right answer.

145. Consider the following statements :

- (i) Fatty acids should be a part of human diet.
- (ii) A normal diet should contain 75 gm of fat.
- (iii) The cells of human body can not synthesize any fatty acids.
- (iv) Deficiency diseases develop due to the absence of essential fatty acids.

Which of the above statements is/are correct ?

- (a) (i), (ii) and (iii)
- (b) (i), (ii) and (iv)
- (c) (ii) and (iii) only
- (d) (iii) and (iv) only

R.A.S./R.T.S. (Pre) 2013

Ans. (b)

The cells of the human body can synthesize most of the fatty acids. However only essential fatty acids cannot be synthesized in the body. Thus, essential fatty acids should be a part of human diet. Deficiency diseases develop due to the absence of essential fatty acids. A normal diet of 2200-2400 calories should contain about 75 gm of fat per day.

146. Linseed is a rich source of –

- (a) Vitamin C
- (b) Omega-3 fatty acid
- (c) Essential amino acids

(d) Antioxidants

R.A.S./R.T.S.(Pre) 2012

Ans. (b)

Linseed is a food and fibre crop that is grown in cooler regions of the world. The oil is extracted from its seed, known as linseed oil, which is a very good source of omega-3 fatty acid, which is able to prevent the deposition of fat in blood vessels.

147. Which among the following is the richest source of alpha-linolenic acid (18- carbon omega-3 fatty acid)?

- (a) Moth bean (b) Oats
(c) Mung bean (d) Flaxseed

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

Alpha-linolenic acid is a kind of omega-3 fatty acid found in plants. It is found in flaxseed oil, canola, soyabean and walnut oil.

148. Assertion (A) : Vitamins are invigorating elements, whose small amount is required for good health.

Reason (R) : Vitamins helps to keep the body healthy by killing harmful bacteria.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but R is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P. Lower Sub. (Pre) 2002

Ans. (c)

Vitamins and minerals are often called micronutrients because our body needs only tiny amounts of them. Cereals, green vegetables, milk, fruit, and meat are a good source of vitamins. White blood cells (not vitamins) help to keeps the body healthy by destroying harmful bacteria.

149. Assertion (A) : Human diet should compulsorily contain glycine, serine and tyrosin.

Reason (R) : Essential amino acids can not be synthesized in the human body.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2002

Ans. (d)

Glycine, serine and tyrosine are non-essential amino acids, which are synthesized inside our bodies. There are 20 biologically active amino acids found in human body. These are divided into two categories – 10 are essential amino acids and 10 are non-essential amino acids. The amino acids, which can be synthesised in the human body, are known as non-essential amino acids. Essential amino acids are those which can't be synthesized in the human body and must be obtained through diet. Hence, (A) is false, but (R) is true.

150. Which one of the following food materials contains all essential amino acids?

- (a) Rice (b) Milk
(c) Egg (d) Soyabean

U.P. P.C.S. (Pre) 2016

Ans. (*)

Amino acids are the building blocks of protein. Foods that contain all essential amino acids are called complete protein foods. Complete proteins are supplied by meat, poultry, yogurt, eggs, fish, milk, quinoa and soyabean etc. Among the given options, except rice, all other food materials contain all essential amino acids.

151. Which of the following eatables is the best source of all the essential amino acids?

- (a) Banana (b) Pulses
(c) Egg (d) Fish

U.P. R.O./A.R.O. (Pre) 2021

Ans. (c)

Egg is the best source of all the essential amino acids for human nutrition because it has the highest Biological Value (BV). The Biological Value (BV) is a scale of measurement used to determine what percentage of a given nutrient source is utilized by the body. BV (for protein sources) is derived from providing a measure intake of protein, then determining the nitrogen uptake versus nitrogen excretion. When a protein source contains the essential amino acids in a proportion similar to that required by the body, it has a high BV, while when one or more of the essential amino acids are missing or present in low numbers, it has a low BV. The Biological Value of some protein sources is given in the following table :

Protein Source	Biological Value
Whole egg	— 94
Milk	— 90
Quinoa	— 83
Fish	— 76
Beef	— 74
Soyabeans	— 73

152. Fat present below the skin surface in our body, acts as a barrier against :

- (a) Loss of heat from the body
- (b) Loss of essential body fluids
- (c) Loss of salts from the body
- (d) Entry of harmful microorganisms from the environment.

I.A.S. (Pre) 1996

Ans. (a)

Fat present below the skin surface of our body forms a complete barrier against the external environment. It is an energy store and also acts as a thermal insulating layer and prevents loss of heat from the body. It protects the underlying structure of the body from any physical trauma.

153. In which category of food is the most amount of calories per unit –

- (a) Vitamins
- (b) Fats
- (c) Carbohydrates
- (d) Proteins

R.A.S./R.T.S. (Pre) 1996

U.P.P.C.S. (Pre) 1999

Ans. (b)

Fats have highest calorie value per unit. 4.0 Kcal energy is obtained by metabolic oxidation of 1 gm carbohydrate or protein whereas about 9.3 Kcal is received from fats of the same amount. Flora and fauna, both are a source of fats.

154. Which of the following gives maximum energy in metabolic processes?

- (a) Carbohydrates
- (b) Fats
- (c) Proteins
- (d) Minerals

U.P.R.O./A.R.O. (Mains) 2013

Ans. (b)

Fats have the highest energy value since 1g of fat contains 9.3 kcal of energy; one gram of proteins contains about 4.0 kcal and one gram of carbohydrates also contains about 4.0 kcal of energy.

155. Which one of the following groups of diet provides maximum energy to human body?

- (a) Fat, Protein, Vitamin
- (b) Vitamin, Mineral, Water
- (c) Carbohydrate, Fat, Protein
- (d) Carbohydrate, Protein, Vitamin

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

Among the given groups of diet, group of Carbohydrate, Fat and Protein provides maximum energy to human body.

156. The substance which makes up about 80% of the living material in most organisms is :

- (a) Protein
- (b) Fats
- (c) Carbohydrates
- (d) Minerals

U.P.P.C.S. (Pre) 1994

Ans. (a)

Generally, water, an inorganic substance is most abundant in living beings. But in organic compounds of the body (living materials) protein is most abundant because they are one of the building blocks of body tissue and can also serve as a fuel source.

157. Doctors recommend that we should cook our food in oil rather than in vanaspathi ghee because –

- (a) Oil contain unsaturated fats
- (b) Oil contain saturated fats
- (c) Oil is easier to store
- (d) Oil is cheaper

Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

Unsaturated fats include the two heart healthy choices - polyunsaturated and monounsaturated fats. Found primarily in fish, plant oils, seeds and nuts, these 'healthy' fats have been shown to lower blood cholesterol levels and lesser risk of cardiovascular disease. Even though these fats are better than saturated and trans fats, they still have fats and intake should be moderate to maintain the good health.

158. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Sunflower oil is rated as a good quality edible oil.

Reason (R) : It contains high quantities of unsaturated fatty acids.

Select the correct answer from the codes given below:

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

U.P. B.E.O. (Pre) 2019

Ans. (a)

The sunflower (*Helianthus*) is an annual plant in the family Asteraceae, with a large flower head (capitulum). Sunflower oil, extracted from its seeds, is used for cooking. Sunflower oil is rated as a good quality edible oil because it contains high quantities of unsaturated fatty acids which reduce the risk of cardiovascular diseases. Sunflower oil is primarily composed of linoleic acid, a polyunsaturated fat and oleic acid, a monounsaturated fat. It is also a rich source of vitamin E.

159. Which vegetable oil is suitable for heart patients?

- (a) Sunflower oil (b) Groundnut oil
(c) Mustard oil (d) Soyabean oil

M.P.P.C.S. (Pre) 2006

Ans. (a)

Sunflower oil is known best for the heart patients. Sunflower oil is produced from oil type sunflower seeds. The oil is light in taste and appearance and provides more vitamin E than any other vegetable oil. It contains monounsaturated and polyunsaturated fats with low saturated fat levels. The U.S. Surgeon Journal reported "Eating less (saturated) fat is the most important change you can make to decrease your risk of heart disease."

160. Assertion (A) : Unsaturated fats are more reactive compared with the saturated fats.

Reason (R) : Unsaturated fats have only single bonds in their structure.

Code :

- (a) Both (A) and (R) are individually true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2003

Ans. (c)

Assertion (A) is true, but (R) is not correct. A compound having a double bond in their structure are more unstable in comparison to single bond compounds. Unsaturated fats, having double bonds in their structures are more reactive than saturated fats.

161. Consumption of fish is considered to be healthy when compared to the flesh of other animals because fish contains :

- (a) Polyunsaturated fatty acids
(b) Saturated fatty acids
(c) Essential vitamins
(d) More carbohydrates and proteins

I.A.S. (Pre) 1998

Ans. (a)

Consumption of fish is considered to be healthy as compared to the flesh of any other animal because it contains polyunsaturated fatty acids, which can reduce LDL (bad) cholesterol level in our blood and also lower the risk of heart disease and stroke.

162. Trans fat is generally considered to be bad for human health because it lowers the level of –

- (a) HDL (b) LDL
(c) Triglycerides (d) Insulin

U.P.P.C.S. (Mains) 2012

Ans. (a)

HDL cholesterol is the well-behaved 'good cholesterol'. This friendly scavenger cruises the blood stream. As it does, it removes harmful bad cholesterol from where it doesn't belong. High HDL levels reduce the risk for heart disease - but low levels increase the risk. Trans fat is generally considered to be bad for human health because it lowers the level of HDL cholesterol.

163. A company marketing food products advertises that its items do not contain trans-fats. What does this campaign signify to the customers?

- The food products are not made out of hydrogenated oils.
- The food products are not made out of animal fats/oils.
- The oils used are not likely to damage the cardiovascular health of the consumers.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2011

Ans. (d)

Eating trans-fats raises the level of low-density lipoprotein (LDL) or bad cholesterol and lowers the level of HDL or good cholesterol, which increases the risk of developing cardiovascular diseases. The animal fats/oils and hydrogenated oils are the forms of trans-fat. Thus, all three statements are correct.

164. What is cholesterol among the following?

- (a) Steroid (b) Vitamin
(c) Enzyme (d) Insecticide

U.P. R.O./A.R.O. (Mains) 2021

U.P. P.C.S. (Pre) 2005

Ans. (a)